



ELE101 Communication Systems

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Outline

- Review
 - Assignment submission
- Communication Systems
- Demo with SDR (Software Defined Radio)
- Python 101 cont'd



GitHub Classroom

1. Click on the assignment link
2. Authorize GitHub Classroom
3. Find yourself on the class roll
4. Match correctly, if you make error contact me!
5. Click on the "Accept this assignment" button.
6. Having accepted the assignment, a new repo will be created for you!.
7. Check your e-mail associated with GitHub for any pending invitation! Accept if any!
8. Read the README file to what to do!



Assignment Submission

- Clone assignment repo into your PC
- change to your local directory (using `cd` command)
- Follow instructions in the README file
- Submit your assignment using git commands
 - `git add .`
 - `git commit -m "something here"`
 - `git push`



How to use conda environment

- First, launch miniforge prompt
- Then, type
 - *conda env list*
 - *conda activate ELE101*
 - *jupyter-lab or jupyter lab*

- [illegible]



Haberleşme

Bilginin karşı tarafa iletilmesi

Sinyal

- Akustik, borsa hisse, deprem, elektronik,
- Işık, radyo dalgaları: haberleşme/radar

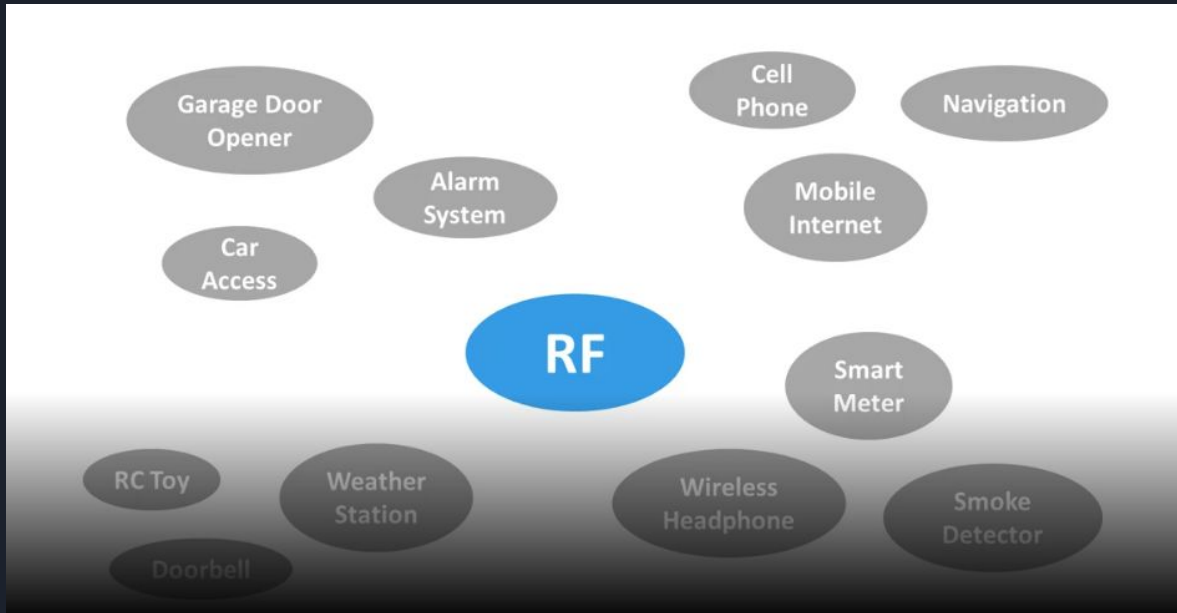
Farklı örnekler

- Mobil el telsizi
- FM Radyo istasyonu
- Nem ölçer

Kanal

- Hava
- Kablo

RF Her Yerde





Understanding Units

- kHz = 1.000 = 10^3 = 10e3
- MHz = 1.000.000 = 10^6 = 10e6
- GHz = 1.000.000.000 = 10^9 = 10e9
- THz = 1.000.000.000.000 = 10^{12} = 10e12

UNITED STATES FREQUENCY ALLOCATIONS

THE RADIO SPECTRUM

RADIO SERVICES COLOR LEGEND

AEROMOBILE MOBILE	MOBILE SATELLITE	RADIO ASTRONOMY
AEROMOBILE FIXED SATELLITE	LAND MOBILE	RADIOCOMMUNICATION SATELLITE
AEROMOBILE RADIOCOMMUNICATION	LAND MOBILE SATELLITE	RADIOLOCATION
AMATEUR	MARITIME MOBILE	RADIOLOCATION SATELLITE
AMATEUR SATELLITE	MARITIME MOBILE SATELLITE	RADIOCOMMUNICATION
BROADCASTING	MARITIME RADIOCOMMUNICATION	RADIOCOMMUNICATION SATELLITE
BROADCASTING SATELLITE	METEOROLOGICAL	SPACE OPERATOR
BROADCASTING SATELLITE	METEOROLOGICAL SATELLITE	SPACE OPERATOR
FOOD	METEOROLOGICAL SATELLITE	SPACE OPERATOR
FOOD SATELLITE	METEOROLOGICAL SATELLITE	SPACE OPERATOR

ACTIVITY CODE

FEDERAL EXCLUSIVE	FEDERAL NON-FEDERAL SHARED
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ALLOCATION USAGE DESIGNATION

SERVICE	CLASS	DESIGNATION
Primary	Fixed	Fixed Station
Secondary	Mobile	Not Licensed with License Condition

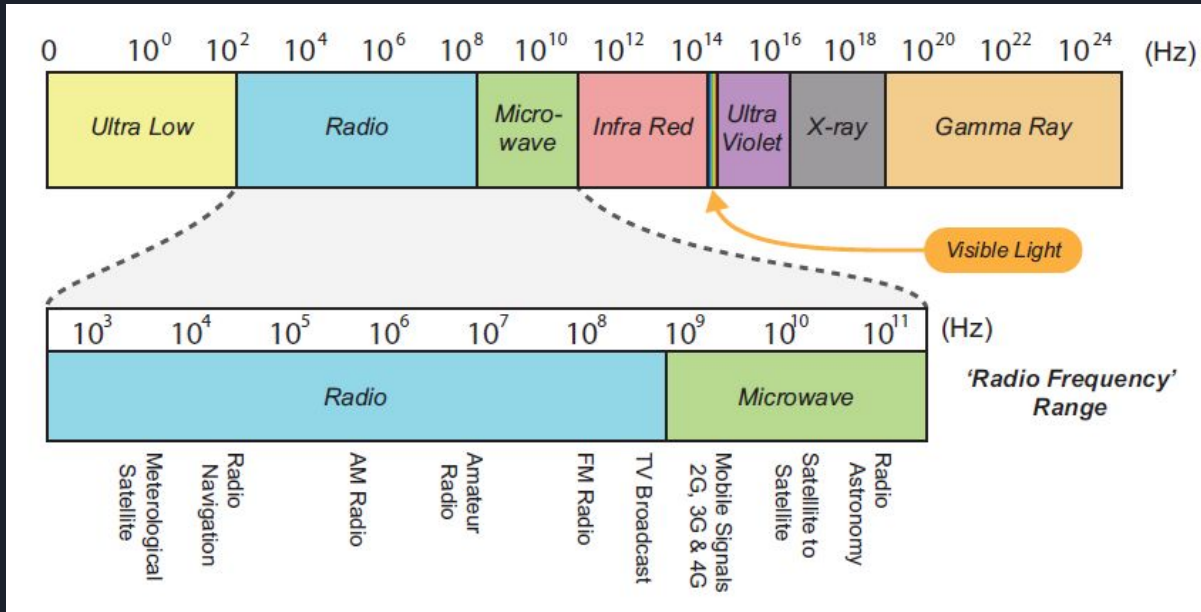
The radio frequency spectrum is managed by the Federal Communications Commission (FCC) and the National Telecommunications and Information Administration (NTIA). The FCC is responsible for the allocation of radio frequency spectrum and the NTIA is responsible for the management of the spectrum. The spectrum is divided into various bands and frequencies, each with specific allocations for different services. The spectrum is a finite resource and its efficient use is crucial for the functioning of modern society. The FCC and NTIA work together to ensure that the spectrum is used in a fair and equitable manner, while also promoting innovation and economic growth. The spectrum is a vital part of our infrastructure and its management is a complex task that requires the expertise of both agencies. The FCC and NTIA are committed to ensuring that the spectrum is used in a way that benefits the public and supports the needs of the economy. The spectrum is a shared resource and its management is a responsibility that we all share. The FCC and NTIA are working to ensure that the spectrum is used in a way that is fair, equitable, and efficient. The spectrum is a vital part of our infrastructure and its management is a complex task that requires the expertise of both agencies. The FCC and NTIA are committed to ensuring that the spectrum is used in a way that benefits the public and supports the needs of the economy. The spectrum is a shared resource and its management is a responsibility that we all share. The FCC and NTIA are working to ensure that the spectrum is used in a way that is fair, equitable, and efficient.

U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management
JANUARY 2016



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Elektromanyetik Spektrum



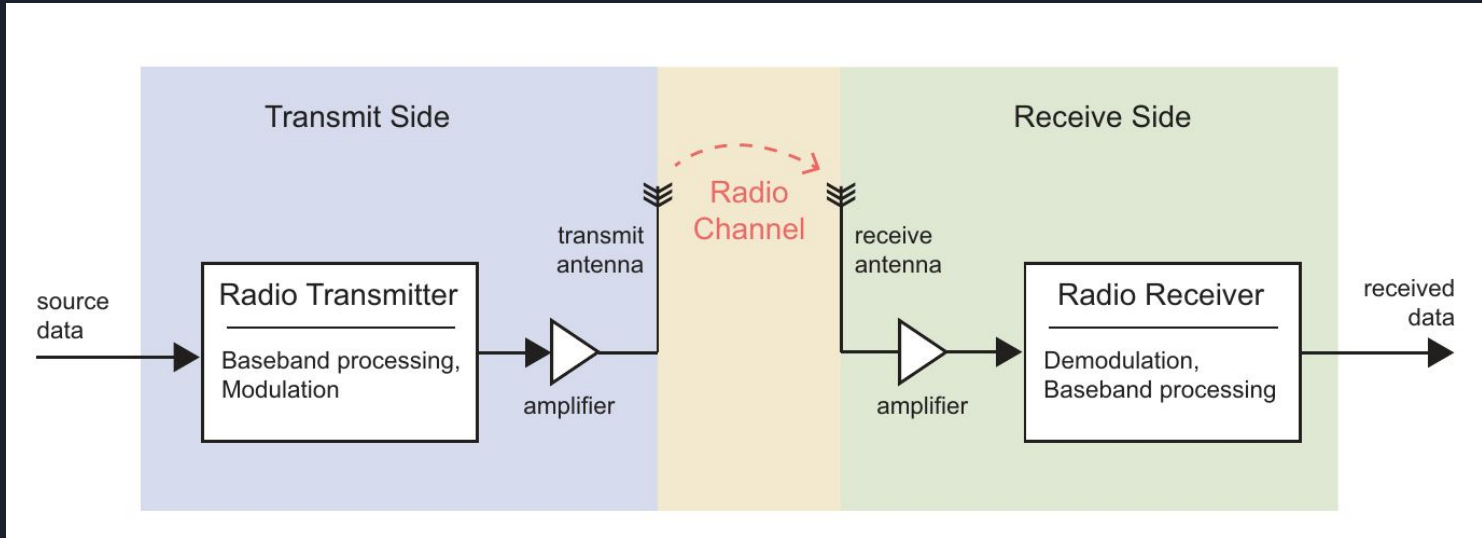
The Electromagnetic Spectrum - NASA



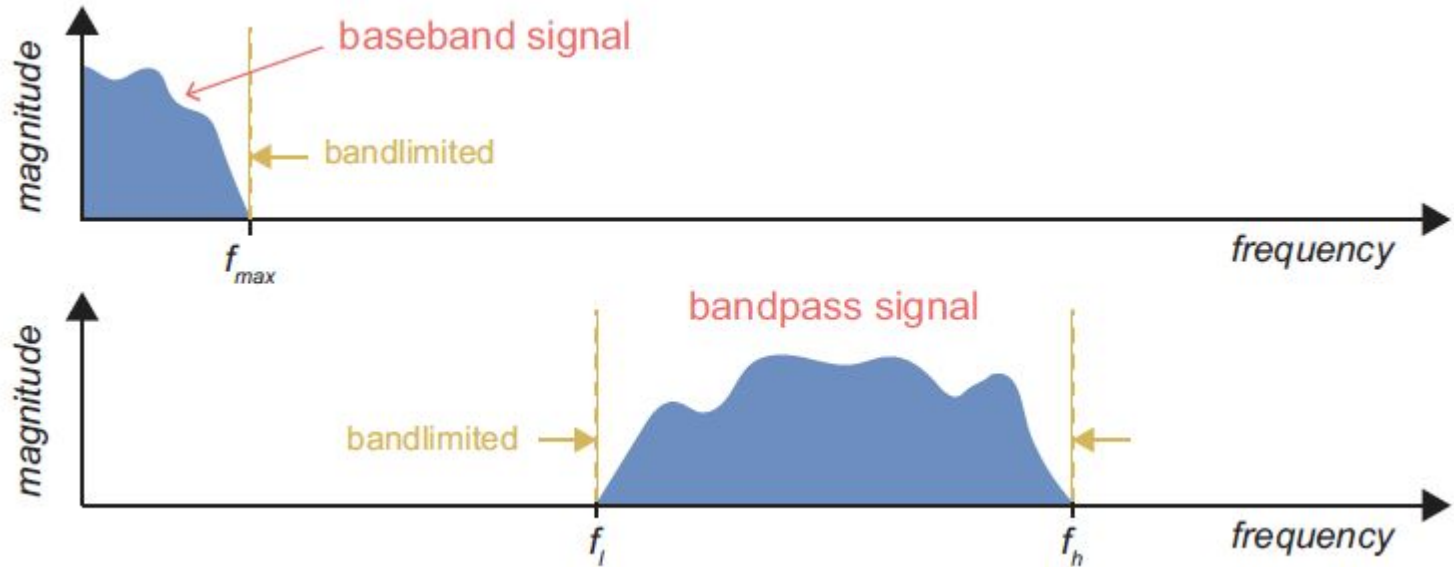
1. Watch the video
2. Answer all quiz questions!

<https://www.flexiquiz.com/SC/N/EMspectrum>

Basit Haberleşme Modeli

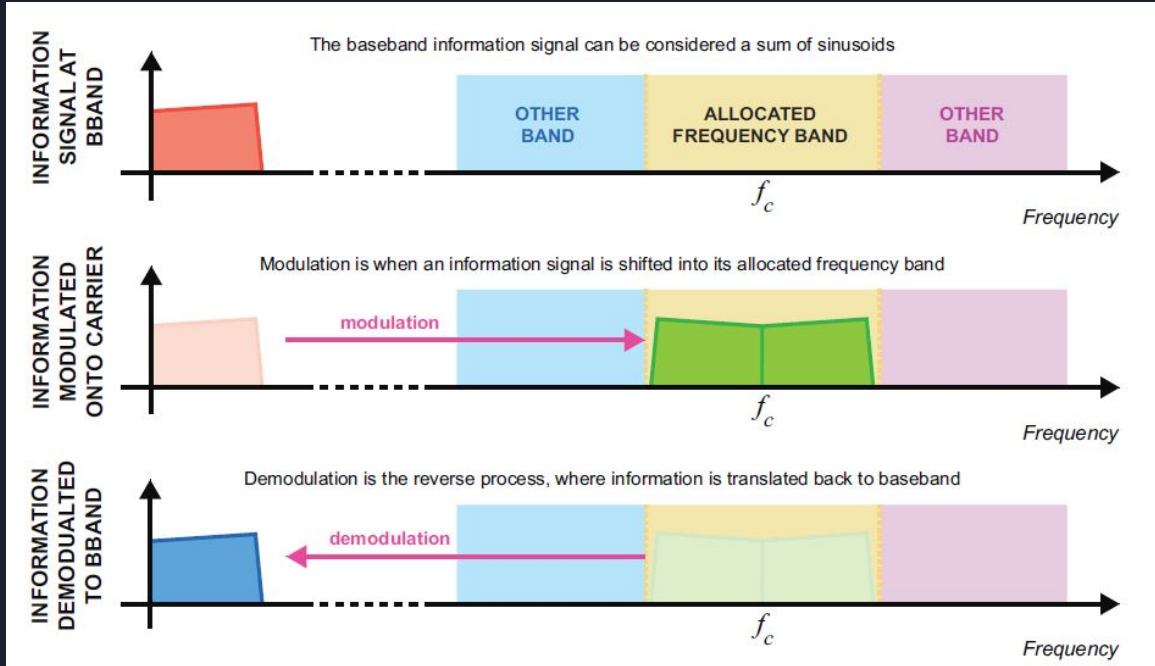


Frekans & Bant Geniřlięi



Modülasyon/Demodülasyon

MoDem = Modülatör + Demodülatör





Communication Systems

- Described as Band, Frequency, or Wavelength
- Bands: VHF
- Frequency: 100 MHz
- Wavelength: 3 m
- $\lambda * \text{frequency} = \text{speed of light}$
 - $\lambda * f = c$
- Wavelength (in m) = $300 / (\text{frequency in MHz})$



FM Radio Example

- What is the wavelength of an FM radio signal at 100 MHz?
- Remember the formula
 - $\text{Wavelength (in m)} = 300 / (\text{frequency in MHz})$
- $300/100 = 3$ meters



5G mmWave Calculation

- 5G uses millimeter waves (mmWaves)
- mmWaves are broadcast at frequencies between 30 and 300 GHz
- So, what are the corresponding wavelengths?



Modülasyon Örnekleri

Analog

- AM
- FM
- PM

Sayısal

- OOK
- FSK
- QPSK/QAM16 ...

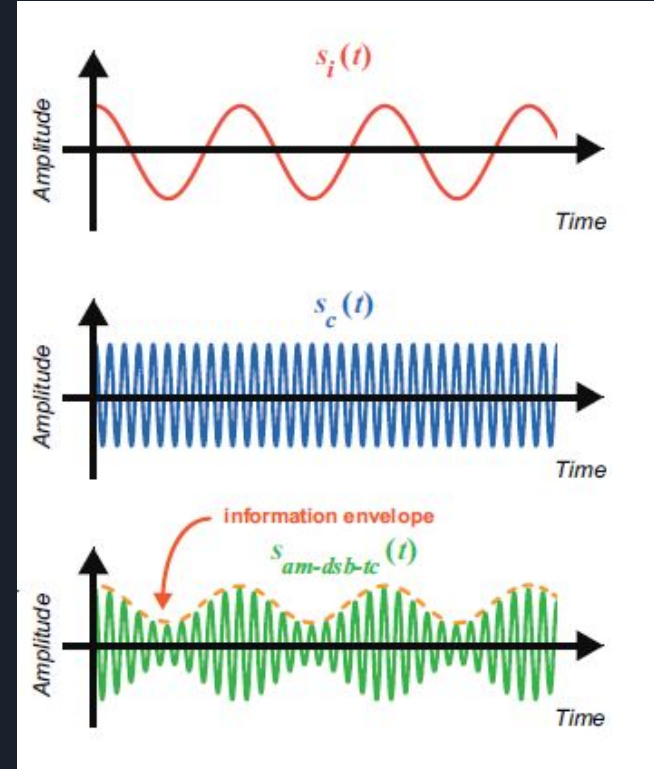
Amplitude Modulation (AM)

Genlik Kiplemesi

Taşıyıcı frekansın genliği değiştiriliyor.

Kompleks örneklenmiş verinin genliği bize bilgiyi veriyor.

Hava bandı konuşmaları



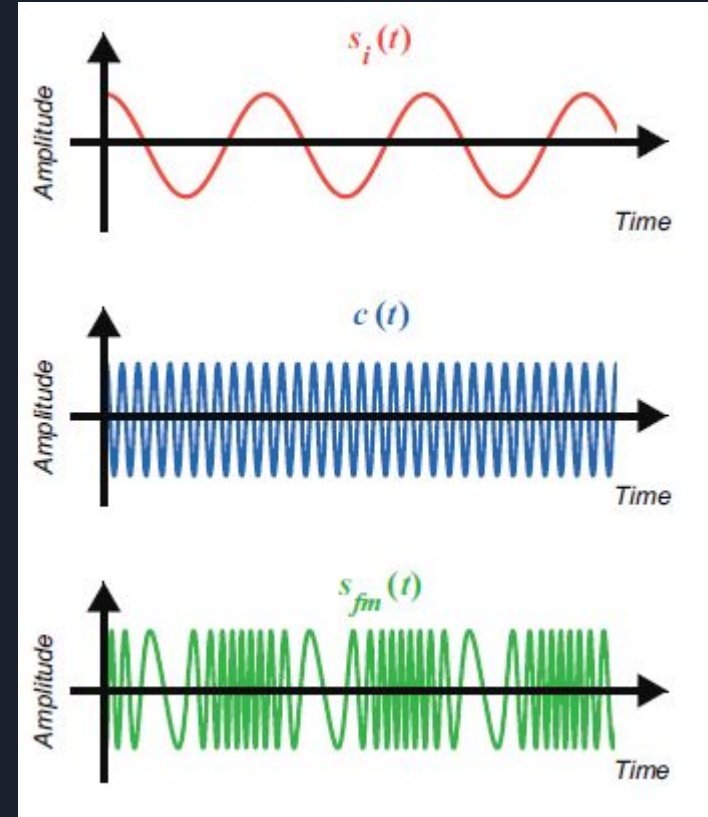
Frequency Modulation (FM)

Frekans Kiplemesi

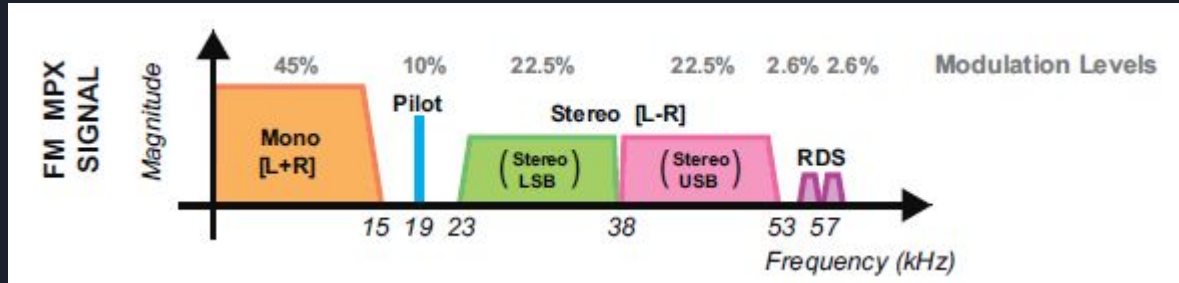
Taşıyıcı frekansın frekansı değiştiriliyor.

Kompleks örneklenmiş verinin türevi bize bilgiyi veriyor.

Gürültüye karşı daha dayanıklı



Geniş Bant Frekans Modülasyonu (WBFM)

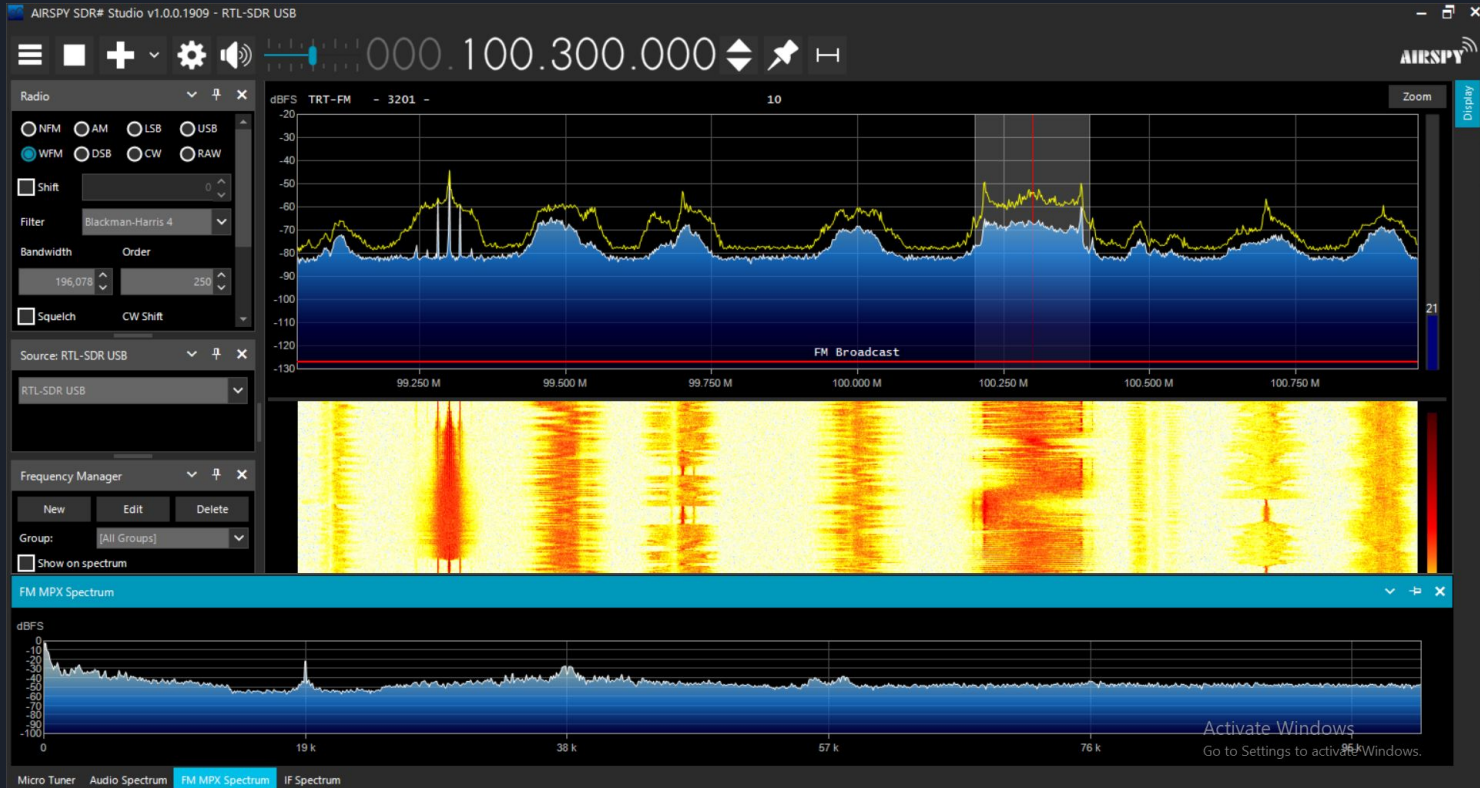




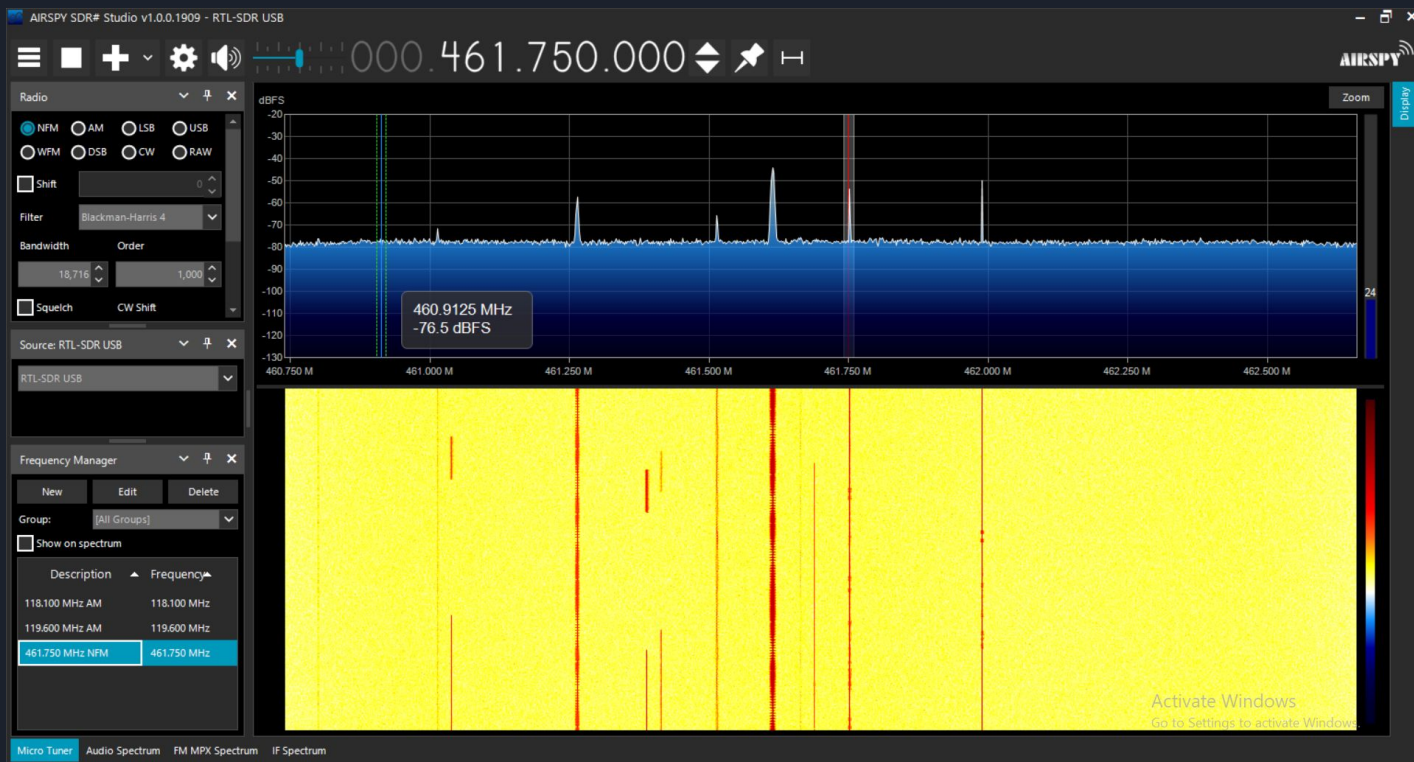
Yazılım Tabanlı Radyo

- Yazılım Tabanlı Radyo (YTR)
- Software Defined Radio (SDR) veya Software Radio (SR)
 - Systems Defined Radio
 - Elektromanyetik, RF, Haberleşme, DSP, Embedded, SW
- Nedir? (IEEE, Wireless Innovation Forum)
 - Bazı veya tüm fiziksel seviye fonksiyonları yazılım ile kontrol edilebilen radyo

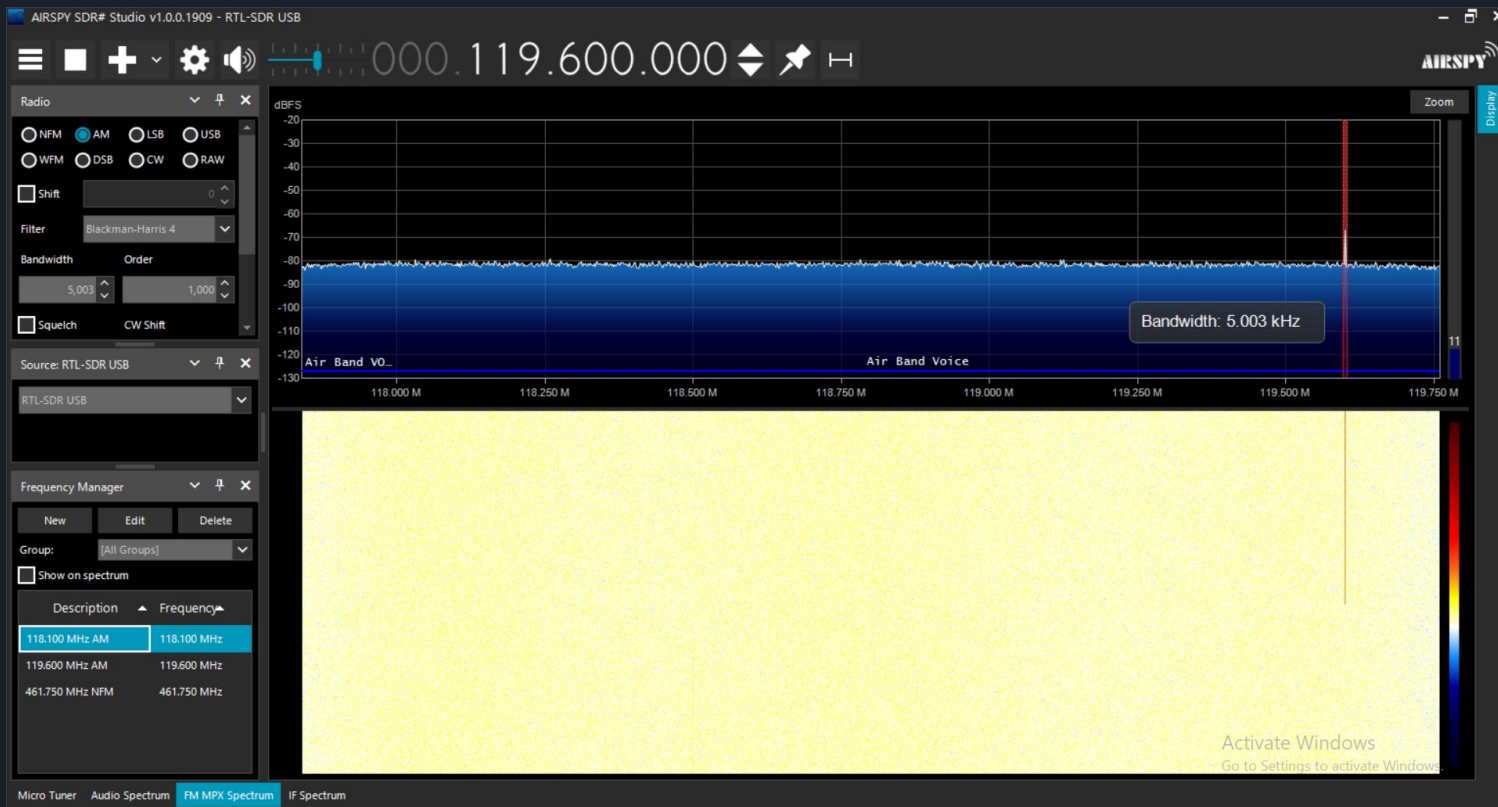
Merhaba Dünya: WBFM



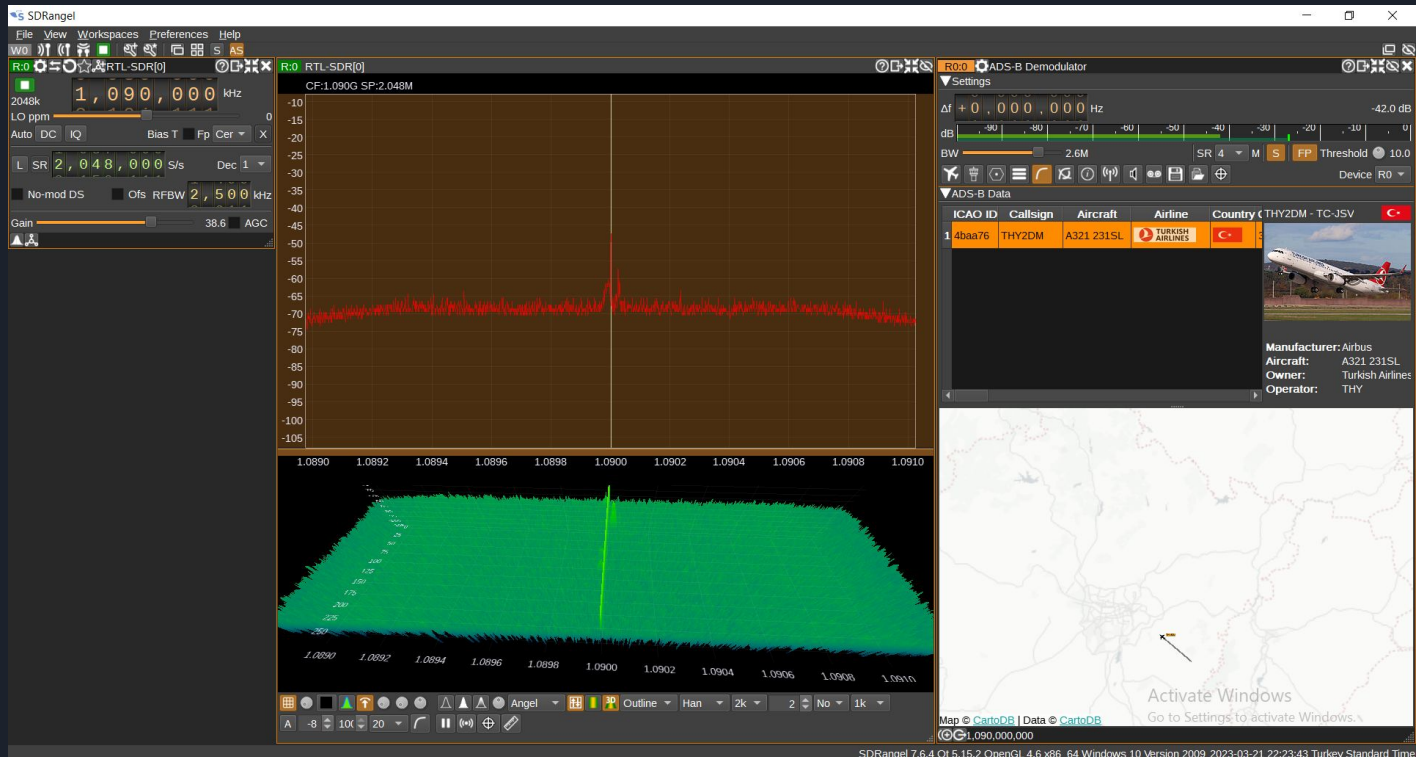
NBFM



AM



ADS-B





Python101

- Open the notebooks



Thanks!

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